PS1
OPENROCKET EXTRATERRESTRIAL FLIGHT DESIGN CHALLENGE

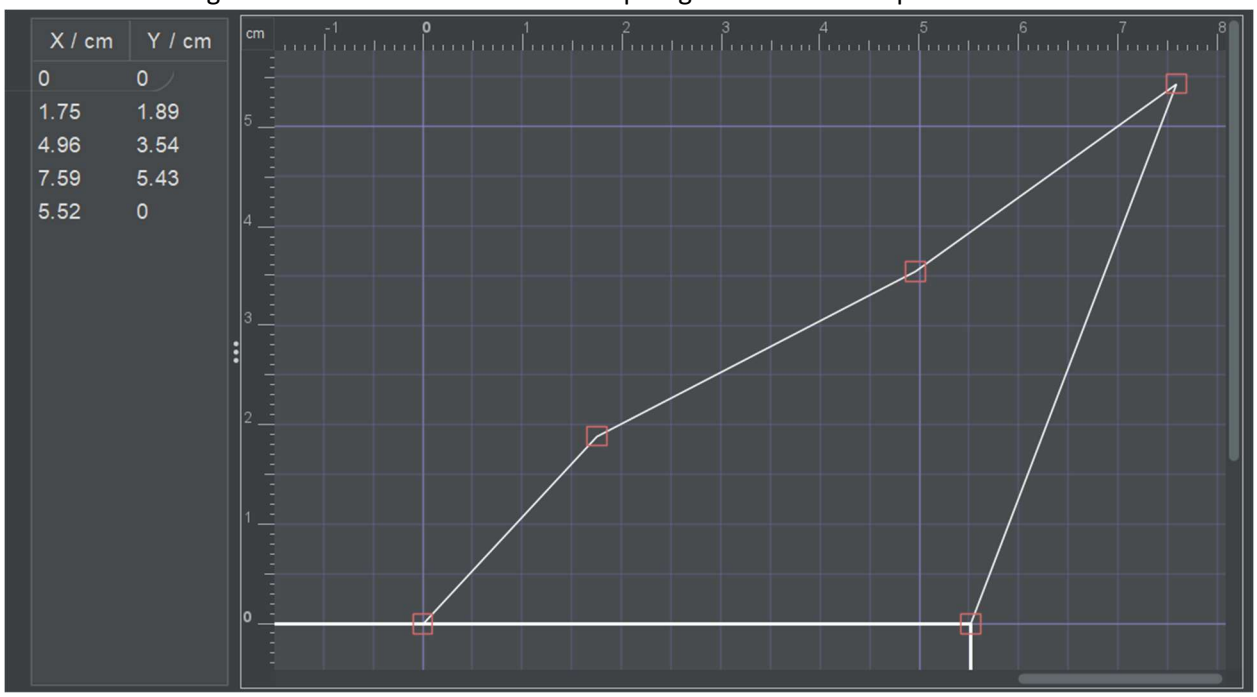
Team- Ball Busters

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1. Design approach and thought process

Our primary thought process during this project was to maximize efficiency, which led us to using cardboard. We played and learnt how changing the length of the tubes, there diameters and adding other supporting structures like the centering ring, tube coupler etc played with out apogee and also the stability.

We share an example where even the fins that we have used are of custom shapes which have been maximized through continuous trial and error to help us get to an ideal shape.



2. Final design summary

Item	Value
Total length	945 mm (limit 2200)
Total liftoff mass incl. motors	2980 g (limit 10)
Payload mass	1 kg (0.95–1.05)
Booster motor	370 N·s
Sustainer motor	858 N·s
Combined total impulse	1228 N·s (limit 2800)
Body tube material	Cardboard
Fins	Custom Shape
Stability margin (launch / minimum in flight)	4.32 cal

3. Recovery system

Device	Diameter	Cd	Deploys	Descends at
Drogue	90	1	Apogee	~ 4 m/s
Main chute	120	1	Apogee (4000 m)	~4 m/s
Payload chute	120	1	Apogee (4000 m)	~4 m/s

[State the descent rate of every branch (payload, sustainer, booster) and the ground hit velocity reported by OpenRocket. Say whether the ground hit figure includes the 10 m/s wind drift.]

4. Mustafar launch conditions

Parameter	Mustafar	Earth reference
Gravity	6.50 m/s²	9.81 m/s²
Pressure	75 kPa	101.3 kPa
Temperature	700 K	288.15 K (15 °C)
Air density	0.374 kg/m³	1.225 kg/m³
Wind	10 m/s at 270° to the launch direction	-

5. Conclusion

While approaching the problem we initially decided to maximize efficiency, i.e less weight, less power required! We chose cardboard because it gives the lightest body structure of the three allowed materials. With total impulse capped at 2800 N·s, every kilogram saved goes straight into apogee. OpenRocket's cardboard has a density of 680 kg/m³, roughly half that of PVC and a quarter that of aluminium.

Similar to how we experimented with the fin design, we also played with Mustafa's low gravity and low air density, all of this helped us greatly understand how to model and configure parts of our rocket, for example requiring a larger parachute but a smaller propulsion system compared to on Earth.

Something I would like to point out is that, during the flight simulations, OpenRocket gives back the "Ground Hit Velocity" which is the net velocity of the object, on plotting the vertical velocity we realise that the **velocity of descent** is actually in the required range.

